Introduction to Artificial Intelligence
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Assignment:
NLP – Question Answering

1 Simple question answering

In this lab session, you will implement a simple NLP question answering solution with a FAQ list as knowledge base (training data). The topic is the movie “A Beautiful Mind”. You will be working in your teams of two students each.

1. Answer the quick check questions
2. Download QA_train_data.csv and QA_test_data.csv from the Moodle course
3. In your Pycharm project, create a new Python package “05b_NLP_QA”
4. Create a new subfolder data and store both CSV files
5. Create a new Jupyter notebook QA_A_Beautiful_Mind.ipynp
6. Implement a simple QA system:
    a. Load data
    b. Implement a predictor
    c. Predict answers for questions in the test set
    d. Use the code provided in the lecture for evaluating the prediction score.
    e. Optimize your solution to potentially increase the score
7. Commit your code to the Git repository
8. Enter your prediction score in the leaderboard in the shared spreadsheet.

2 Optional: Full question answering

Alternatively, you may implement a QA system without using the FAQ list as knowledge base (training data). Only use the text about the movie “A Beautiful Mind” provided in the lecture (respectively from Wikipedia). Use QA_test_data.csv and the code provided in the lecture for evaluating the prediction score. Enter your prediction score in the leaderboard in the shared spreadsheet with a comment that a full QA solution was implemented.